

ANALYSTLAB AFRICA

Experience Lab Internship Programme

HealthConnect Clinic – Machine Learning Engineering Track

Week 4: ML System Design Document

Programme	AnalystLab Africa Experience Lab
Week	Week 4 – Problem Understanding & System Design
Project	HealthConnect Clinic Experience Lab
Project Title	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Track	Machine Learning Engineering
Document	ML System Design Document
Version	1.0
Focus	Understanding, Planning & Initial System Design (no model training required)

1. ML Problem Overview

1.1 Business Context

HealthConnect Clinic is a fictional outpatient healthcare provider that manages appointment-based services. The clinic faces significant operational challenges linked to patient no-shows, including inefficient use of appointment slots, reduced clinic capacity, longer waiting times for other patients, and increased administrative overhead from last-minute cancellations and rescheduling.

Central Project Question: How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

1.2 Machine Learning Engineering Objective

The responsibility of the Machine Learning Engineering track is to design a reliable, reproducible, and production-oriented machine learning system capable of predicting the likelihood that a patient will miss (no-show) a scheduled appointment.

The system must:

- Produce predictions that can support operational decision-making (targeted reminders, overbooking strategies, prioritised outreach).
- Be structured for reproducibility, version control, and future model management.
- Integrate cleanly with potential downstream systems (dashboards, notification services, or GenAI assistants).

Week 4 focus: System design and planning only. No model training, deployment, or final implementation is expected at this stage.

1.3 Success Criteria (High-Level)

- Ability to generate a risk score (probability of no-show) for each upcoming appointment.
- Clear and documented input/output contract.
- Versioned pipeline that can be re-run and audited.
- Early identification of key dependencies and risks.

2. Input and Output Definition

2.1 System Inputs

The prediction system will consume structured appointment records. Based on the provided Data Dictionary, the expected input features are summarised below.

Feature	Type	Description	Notes
patient_id	Text	Anonymised patient identifier	Used for historical aggregation
gender	Text	Recorded gender category	Categorical
age / age_group	Integer / Text	Patient age and derived band	Adult patients only
appointment_type	Text	Type of appointment	Four categories
booking_lead_days	Integer	Days between booking and appointment	Key behavioural signal
previous_appointments	Integer	Historical appointment count	Patient history
previous_no_shows	Integer	Historical no-show count	Strong predictor
reminder_sent	Text	Whether a reminder was sent (Yes/No)	Binary
reminder_channel	Text	Channel used (SMS, etc.) or None	Categorical
distance_to_clinic_km	Decimal	Approximate distance to clinic	Limited missing values
waiting_time_minutes	Integer	Estimated waiting time	Limited missing values
appointment_day	Text	Day of the week	Derived
appointment_time	Text	Time period (Morning/Afternoon/Evening)	Categorical
appointment_date / booking_date	Date	Temporal features	Derived features possible

Input format: Structured tabular data (CSV or equivalent), one row per appointment. Scoring mode: Initially designed for batch scoring of upcoming appointments (daily or weekly).

2.2 System Outputs

Output Field	Type	Description
appointment_id	Text	Unique identifier (pass-through)
no_show_probability	Float (0-1)	Predicted probability of no-show
no_show_risk_label	Text	Risk category (Low / Medium / High) – optional threshold-based
prediction_timestamp	Datetime	When the prediction was generated
model_version	Text	Version of the model used

Primary target variable: Binary classification of appointment_outcome where Positive class = No-Show and Negative class = Attended. Cancelled appointments require explicit handling (see section 4.3).

3. High-Level System Architecture

The architecture follows a modular, layered design that separates data handling, feature engineering, model training, versioning, and inference. This supports reproducibility and future evolution of the system.

1. Data Sources

HealthConnect_Appointment_Data.csv + Data Dictionary

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2. Data Ingestion & Validation Layer

Schema validation • Missing value detection • Basic data quality checks

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3. Feature Engineering & Preprocessing

Handling of cancelled appointments • Encoding • Imputation • Feature derivation • Train/Val/Test split (temporal awareness)

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4. Model Training Pipeline

Candidate models (Logistic Regression, Random Forest, Gradient Boosting) • Hyperparameter tuning • Cross-validation • Model selection

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5. Model Registry & Versioning

Model artefacts (joblib / ONNX) • Metadata (metrics, features, training date, data hash) • Version control (Git + model registry)

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6. Inference / Scoring Service

Batch scoring of upcoming appointments • Optional REST API (future) • Output: probability + risk label + model version

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7. Downstream Consumers

Analytics / BI dashboards • Operational alerts / targeted reminder system • Integration with Generative AI assistant (future)

Key design principles: Separation of concerns (data, features, model, serving); Reproducibility by design (seed control, data versioning, config files); Observability (logging of data quality, prediction distributions, drift indicators).

4. Proposed ML Workflow

4.1 End-to-End Stages

1. Data Acquisition & Understanding

Load HealthConnect_Appointment_Data.csv; cross-reference with Data Dictionary; explore distribution of appointment_outcome.

2. Data Quality Assessment & Cleaning

Check missing values (distance_to_clinic_km, waiting_time_minutes); validate consistency (previous_no_shows ≤ previous_appointments); detect outliers.

3. Target Definition & Handling of Cancellations

Primary binary target: No-Show vs Attended. Cancelled appointments: recommended Option A – exclude from training set for initial prototype; Option B – multi-class later. Decision must be documented.

4. Feature Engineering

Patient-level aggregates (historical no-show rate); temporal features (day of week, lead-time bins); optional interaction features.

5. Train / Validation / Test Split

Prefer temporal split (train on earlier appointments, test on later ones) to simulate real-world deployment and avoid leakage.

6. Model Development

Baseline: Logistic Regression. Stronger candidates: Random Forest, XGBoost / LightGBM. Metrics: AUC-ROC, Precision-Recall, F1, Calibration (Brier score).

7. Model Evaluation & Selection

Business-oriented thresholds (e.g. high-risk patients for extra outreach); fairness checks across age groups / gender if relevant.

8. Packaging & Versioning

Save model + preprocessing pipeline as a single artefact; record metrics, feature list, data snapshot hash, and code version.

9. Inference Pipeline Design

Batch scoring script; logging of predictions for monitoring.

10. Monitoring & Retraining Strategy

Data drift detection; performance degradation alerts; periodic retraining plan (high-level for Week 4).

4.2 Reproducibility, Version Control & Model Management

Aspect	Approach
Code versioning	Git + GitHub repository with clear folder structure
Data versioning	Keep original CSV immutable; store cleaned versions with timestamps/hashes
Experiment tracking	Simple config files + results logging (or MLflow later)
Model artefacts	Versioned folders (models/v1/, models/v2/) + metadata JSON
Environment	requirements.txt or environment.yml + fixed random seeds
Documentation	README + this System Design Document + experiment notes

5. Key Dependencies

5.1 Data Dependencies

- Availability and quality of HealthConnect_Appointment_Data.csv
- Completeness of key predictors (previous_no_shows, booking_lead_days, reminder_sent)
- Limited missing values in distance_to_clinic_km and waiting_time_minutes
- Synthetic nature of the data (must be treated as realistic but not real-world)

5.2 Technical Dependencies

- Python ecosystem (pandas, scikit-learn; optionally XGBoost/LightGBM)
- Jupyter Notebook or VS Code for development
- Git & GitHub for version control
- Sufficient compute for training (local or free-tier Colab adequate for this dataset size)

5.3 System & Organisational Dependencies

- Clear definition of how predictions will be consumed (Analytics track, operations team, GenAI assistant)
- Agreement on target definition (especially handling of Cancelled)
- Future integration points with reminder systems or clinic operations
- Ethical / fairness review before any operational use

5.4 Cross-Track Dependencies

- Data Analytics / Data Science tracks: Shared understanding of data quality and feature importance.
- Generative AI track: Potential future use of no-show risk scores to personalise assistant responses or trigger escalations.
- Project Management track: Alignment on timeline and deliverables.

6. Initial Implementation Approach

6.1 Recommended Project Structure (GitHub)

```
healthconnect-ml-engineering/  
├── data/  
│   ├── raw/ # Original dataset (immutable)  
│   └── processed/ # Cleaned & feature-engineered datasets  
├── notebooks/  
│   ├── 01_data_exploration.ipynb  
│   ├── 02_feature_engineering.ipynb  
│   └── 03_model_prototyping.ipynb  
├── src/  
│   ├── data_validation.py  
│   ├── feature_engineering.py  
│   ├── train.py  
│   ├── predict.py  
│   └── utils.py  
├── models/ # Versioned model artefacts  
├── configs/ # YAML/JSON configs  
├── tests/  
├── docs/  
│   └── ML_System_Design_Document.md  
├── requirements.txt  
└── README.md
```

6.2 Week 4 → Week 5 Roadmap (High-Level)

- Week 4 (current): Problem understanding, system design, data assessment, target definition, architecture.
- Week 5 (proposed): Data cleaning, feature engineering, baseline model, initial evaluation metrics, documentation of findings.

6.3 Tools & Libraries (Initial Set)

- Python 3.10+
- pandas, numpy
- scikit-learn
- matplotlib / seaborn (exploratory analysis)
- joblib or pickle for model serialisation
- Git + GitHub
- Optional later: XGBoost / LightGBM, MLflow, FastAPI (for serving)

7. Assumptions, Limitations, Risks and Dependencies

7.1 Assumptions

- The dataset is representative enough of real appointment behaviour for prototype purposes.
- appointment_outcome is correctly labelled.
- Historical features (previous_no_shows, previous_appointments) are available at prediction time.
- Clinic operations can act on risk scores (targeted reminders, overbooking, etc.).
- Cancelled appointments can be reasonably excluded or treated separately without major bias.

7.2 Limitations

- Synthetic / fictional data → results may not generalise to real clinics.
- Limited number of features (no socio-economic, clinical severity, or weather data).
- Missing values in distance and waiting time.
- No explicit patient-level longitudinal modelling yet (simple aggregates only).
- Binary classification may oversimplify multi-outcome reality (Attended / No-Show / Cancelled).

7.3 Risks

Risk	Impact	Mitigation
Data leakage (temporal)	High	Use time-based split
Class imbalance (if No-Show rate is low)	Medium	Stratified sampling, class weights, or PR-AUC
Concept drift over time	Medium	Plan for monitoring and retraining
Misuse of predictions (over-reliance)	High	Clear documentation of model limitations + human-in-the-loop
Privacy / ethical concerns	Medium	Anonymised data already; avoid sensitive inferences
Integration failure with other tracks	Medium	Early alignment on interfaces

7.4 Modelling Considerations

- Calibration of probabilities is important if risk scores are used for prioritisation.
- Threshold selection should be driven by business cost of false positives vs false negatives.
- Feature importance analysis will help explainability and trust.

8. Week 4 Project Summary

1. Specific problem addressed by the ML Engineering track

Design a reliable and reproducible machine learning system to predict patient appointment no-shows so that HealthConnect Clinic can take proactive operational actions.

2. Resources used

- HealthConnect Appointment Dataset (HealthConnect_Appointment_Data.csv)
- HealthConnect Data Dictionary
- Week 4 Assignment document
- (Knowledge Base is primarily for GenAI track; not directly used here)

3. Key observations from Week 4

- The data contains clear candidate predictors (historical no-shows, lead time, reminders, distance).
- Target definition requires careful handling of the “Cancelled” class.
- Limited missing values exist and must be addressed.
- A modular, versioned architecture is essential for reproducibility and future iteration.

4. Proposed approach

Build a modular pipeline (ingestion → feature engineering → training → versioning → batch inference) with strong emphasis on reproducibility, clear input/output contracts, and documented assumptions.

5. Key considerations that may affect the project

- Handling of cancelled appointments
- Temporal leakage risk
- Class imbalance and calibration
- Cross-track alignment on how predictions will be consumed
- Synthetic nature of the data

6. Proposed focus for Week 5

- Perform thorough data quality assessment and cleaning
- Implement feature engineering pipeline
- Train and evaluate a strong baseline model
- Document initial performance metrics and feature importance
- Refine the system design based on empirical findings

9. Next Steps & Professional Development Notes

- Maintain this document and the GitHub repository as living artefacts.
 - Update README.md with Week 4 progress and clear instructions.
 - Document all design decisions and trade-offs.
 - Share professional updates on LinkedIn / X tagging AnalystLab Africa when appropriate.
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Document prepared for: AnalystLab Africa Experience Lab – Week 4
Track: Machine Learning Engineering
Status: Complete for Week 4 requirements

This document provides a solid, organised foundation for the development phase starting in Week 5.